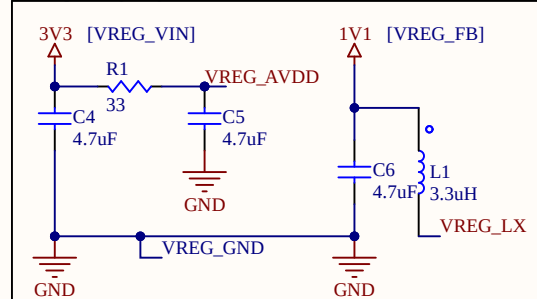
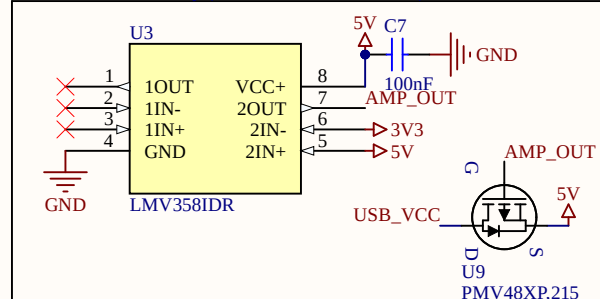


On-chip voltage regulation circuit [3v3 => 1v1]

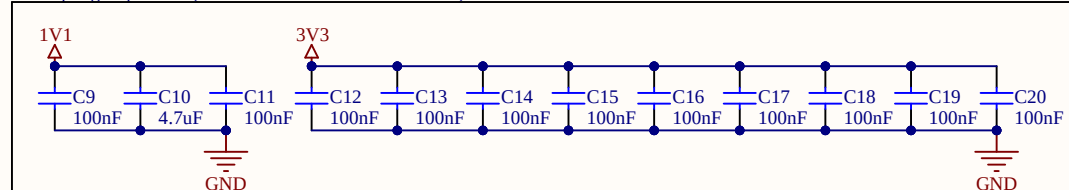


See pg 8 of hardware design manual for PCB layout requirements

Power selection circuitry (choose VIN or USB_VCC)

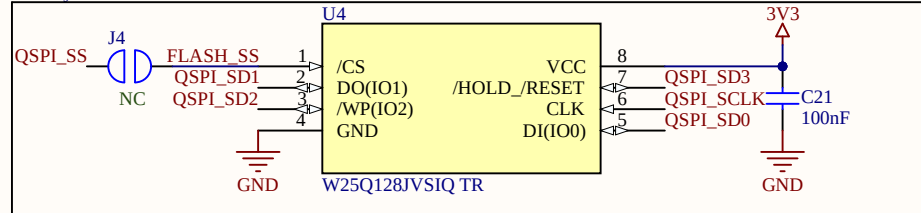


Decoupling Capacitors (for 1V1 DVDD and 3V3 IOVDD)

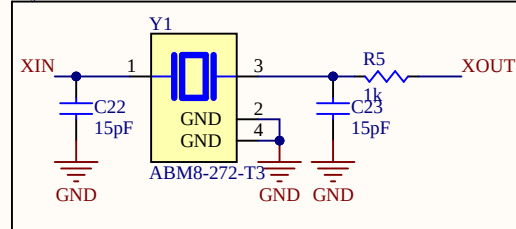


Decoupling caps should be as close to the VDD pins as possible. Because of certain design tradeoffs, some pins may share a decoupling capacitor.

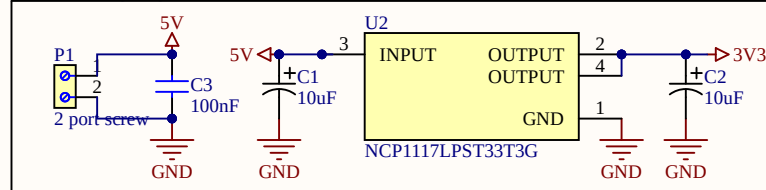
Primary Flash and Boot Button



Crystal Oscillator

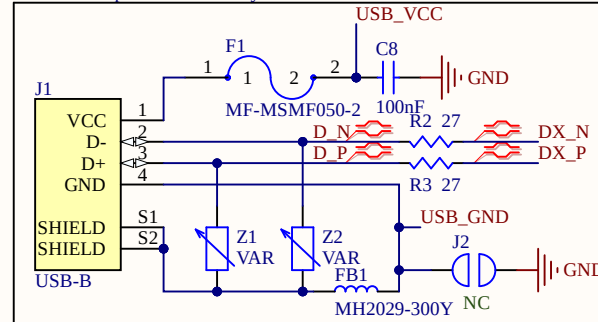


External voltage regulator [5v => 3v3]

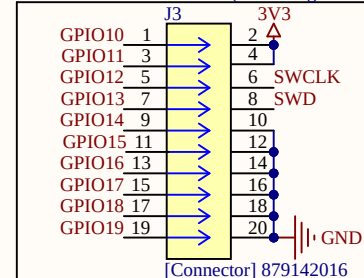


Assume a clean 5V will be provided to the board

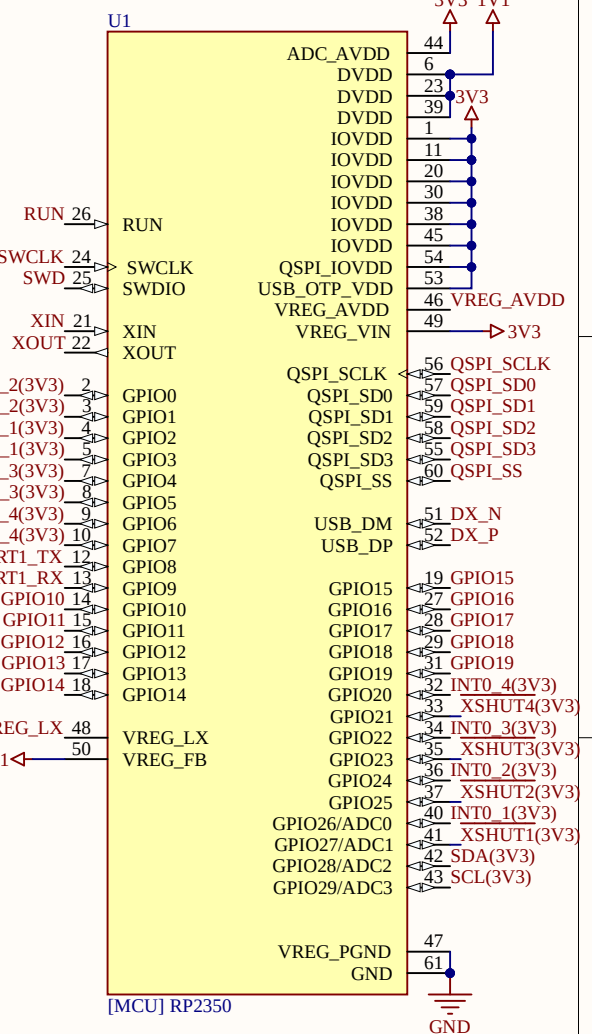
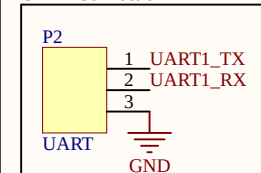
USB-B with protection circuitry



2x10 Extra GPIO Breakout (and debug connections)



UART Connection



Project/Vehicle: Distance Sensor Interface

Author(s):
-Aiden SheplerRevisor(s):
-*MIL
MCU setup according to RPi's
hardware design guidelinesGit Repo: *
Git Hash: 9d107eae

Date: 2/23/2026

Revision: 0

Size: A

File: MCU.SchDoc

Sheet 1 of 2

A

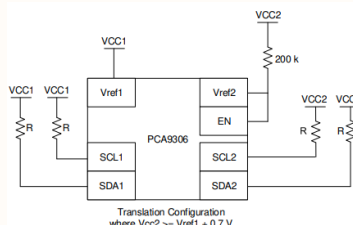
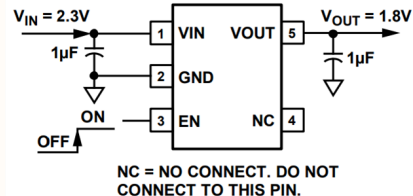
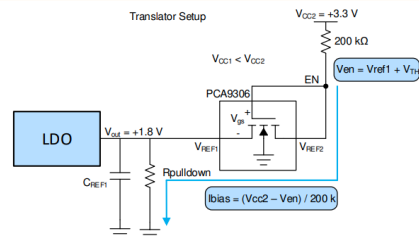


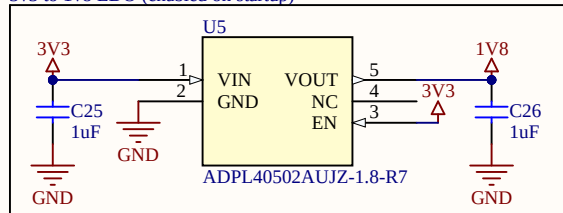
Figure 8-6. Translation Configuration



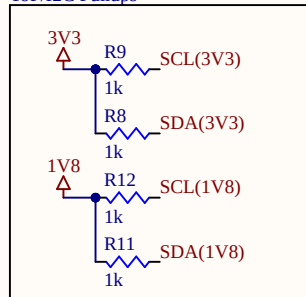
001



3v3 to 1v8 LDO (enabled on startup)

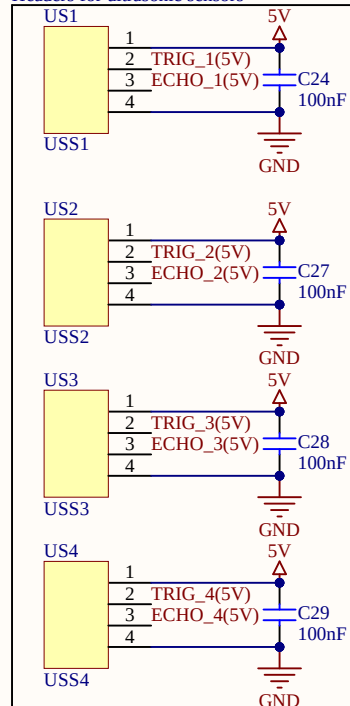


ToF/I2C Pullups

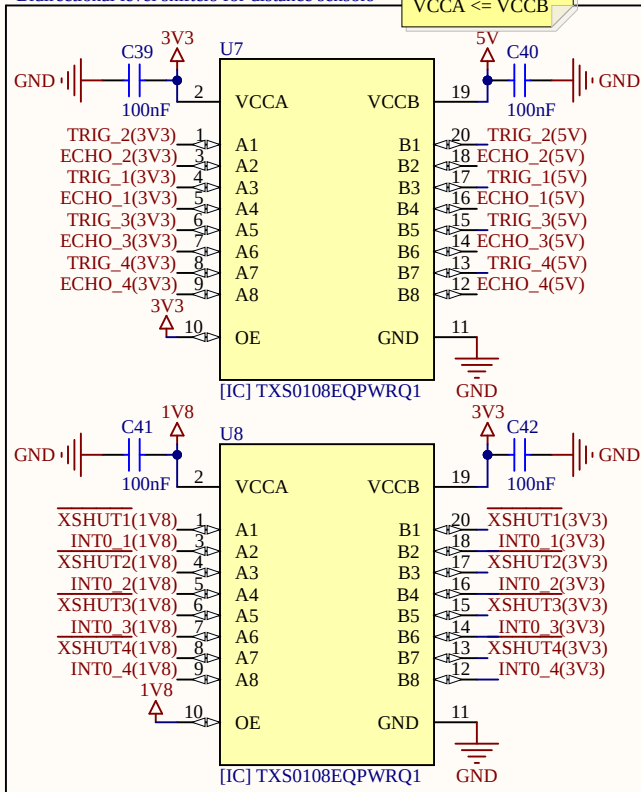


Note: GPIO1/INT0 and XSHUT pins for the ToF are pulled-up on the ToF breakout board

Headers for ultrasonic sensors

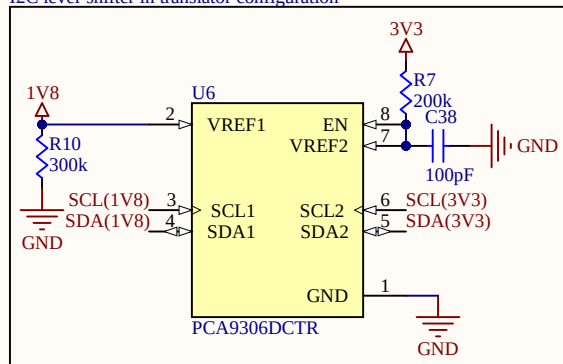


Bidirectional level shifters for distance sensors



B

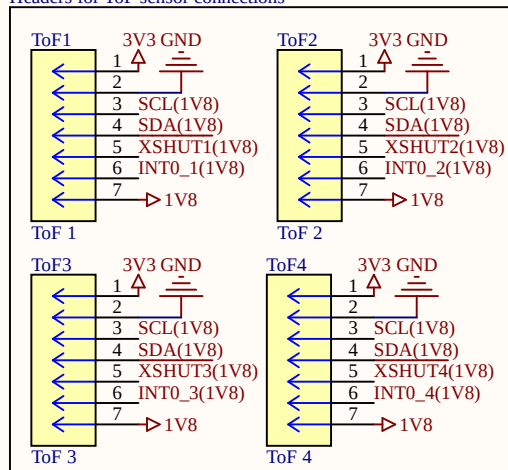
I2C level-shifter in translator configuration



C

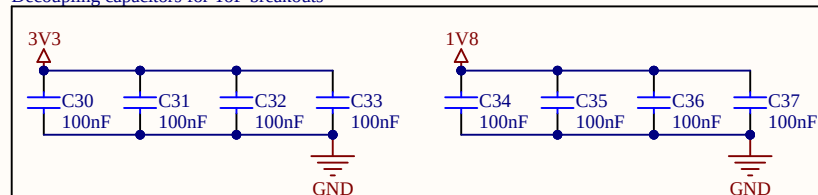
See the wiki page for the pulldown resistor calculation.

Headers for ToF sensor connections



D

Decoupling capacitors for ToF breakouts



Project/Vehicle: Distance Sensor Interface

Author(s):
-Aiden Shepler-
-
-
-

Revisor(s):

-
-
-
-

MIL

Level-shifting circuitry for ToF and headers for both ToF and ultrasonic sensors

Git Repo: *

Git Hash: 9d107eae

Date: 2/23/2026

Revision: 0

Size: A

File: Distance-Sensors.SchDoc

Sheet 2 of 2



Board Stack Report